SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Electronic Devices & Circuits: Diodes, BJTs, MOSFETs & Amplifiers

Department: ECE	**Course / Subject:** Electronic Devices & Circuits (EC301PC)
Semester: Semester 3	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: PN junction semiconductor physics, Zener voltage regulators, Bipolar Junction Transistor (BJT) configurations, JFET, MOSFET operation, and Small-signal models.

Unit 1: Semiconductor Diodes

Carrier concentrations, Drift and diffusion currents, PN junction diode V-I characteristics, Diode capacitance, Half-wave and Full-wave rectifiers with filters.

Unit 2: Special Semiconductor Devices

Zener diode avalanche breakdown, Zener voltage regulator design, Tunnel diode, Varactor diode, Photodiode, Solar cells, and Light Emitting Diodes (LED).

Unit 3: Bipolar Junction Transistors (BJT)

BJT physical structure, Transistor currents, Common Base (CB), Common Emitter (CE), and Common Collector (CC) configurations, Input/Output characteristics, Early effect.

Unit 4: Transistor Biasing & Stabilization

Operating point (Q-point), DC load line, Fixed bias, Collector-to-base bias, Voltage divider bias, Stability factor analysis (S, S', S''), Thermal runaway prevention.

Unit 5: Field Effect Transistors (FET & MOSFET)

JFET construction, Pinch-off voltage, Depletion and Enhancement type MOSFETs, Drain and Transfer characteristics, Small signal h-parameter equivalent circuit models.